

FINAL VALIDATED PERFORMANCE SUMMARY TABLE (Mean ± SD Across Seeds 42, 43, 44, 45, 46)

50 Nodes | 600m River | 2.0 J Initial Energy | 4000-bit Packets | Failure Event at Round 500

Protocol	FND (rnd)	HND (rnd)	LND (rnd)	Final Alive	PDR (%)	Delivered (rep)	Total Energy (J)	Energy/Del (J)	Efficiency (rep/J)
LEACH	23.2 ± 2.3	105.0 ± 36.8	908 (1/5)	6.2 ± 5.8	62.79% ± 5.24%	9260 ± 2968	97.0 ± 2.7	0.0114 ± 0.0039	96.1 ± 33.0
DEEC	26.4 ± 7.9	116.2 ± 46.1	814 (2/5)	3.8 ± 5.0	57.13% ± 3.93%	8018 ± 2835	98.7 ± 1.8	0.0136 ± 0.0050	81.6 ± 30.5
PEGASIS	74.0 ± 71.7	772.0 ± 161.2	>1000	23.4 ± 3.3	29.23% ± 1.53%	10349 ± 542	70.8 ± 5.4	0.0069 ± 0.0008	147.3 ± 18.6
RL-Hybrid	475.4 ± 55.0	645.0 ± 49.2	>1000	14.0 ± 14.9	86.10% ± 0.53%	31889 ± 5318	90.6 ± 12.7	0.0029 ± 0.0007	365.9 ± 129.4

KEY PERFORMANCE FINDINGS (Statistically Supported Across Seeds 42-46):

- **Highest Packet Delivery Ratio (PDR):** RL-Hybrid achieves 86.10% ± 0.53% (vs 62.79% LEACH, 57.13% DEEC, 29.23% PEGASIS).
- **Highest Energy Efficiency:** RL-Hybrid delivers 365.9 ± 129.4 reports/J (vs 96.1 LEACH, 81.6 DEEC, 147.3 PEGASIS).
- **Lowest Energy per Delivered Report:** RL-Hybrid requires 0.0029 ± 0.0007 J/report (vs 0.0114 J LEACH, 0.0136 J DEEC, 0.0069 J PEGASIS).
- **Latest First Node Dead (FND):** RL-Hybrid maintains full 50-node network intact longest (475.4 ± 55.0 rounds vs 23.2 LEACH, 26.4 DEEC, 74.0 PEGASIS).
- **Half Node Dead (HND) Tradeoff:** PEGASIS achieves superior HND (772.0 ± 161.2 rounds) due to aggressive single-packet aggregation, but at severely compromised delivery ratio (29.23% PDR).